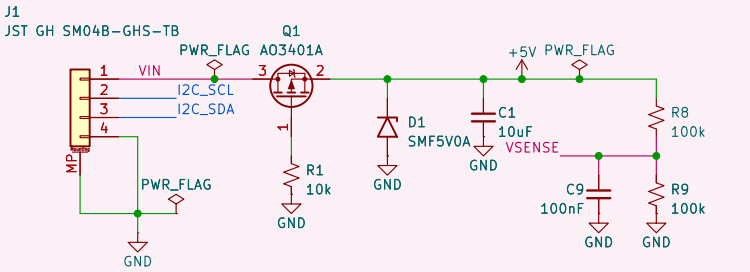
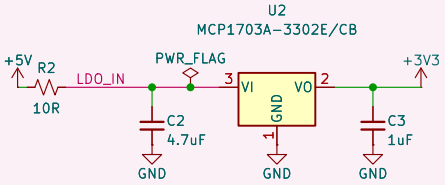
POWER INPUT, PROTECTION, SUPPLY SENSE



VSENSE = +5 V / 2 (PA4)

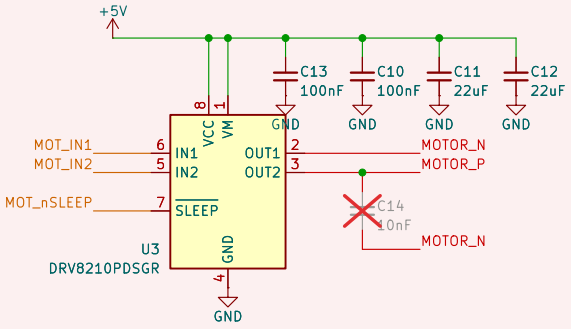
J1: JST GH 1.25 mm, Dronecode/Pixhawk I2C pinout: 1 = +5 V, 2 = SCL, 3 = SDA, 4 = GND

3.3 V LDO



10R + 4.7uF: keeps motor PWM ripple out of the LDO.
MCP1703A: 16 V max in, 2 uA Iq

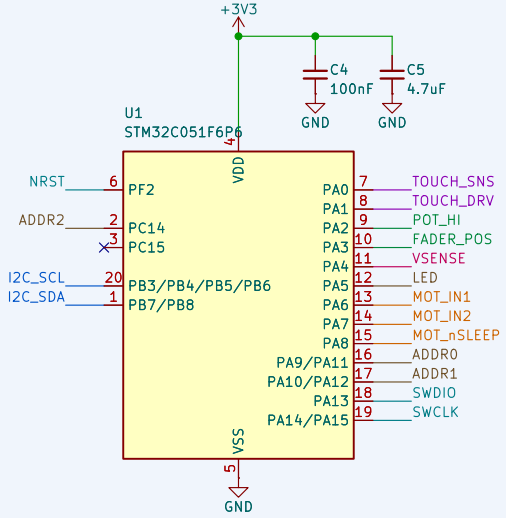
MOTOR DRIVER (DRV8210P, DRV8837-compatible, motor from +5



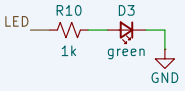
C13 = VCC, C10-C12 = VM decoupling. C14 not fitted (brush-noise option).
IN2 = PWM, IN1 = 0: M+ positive, knob moves to the top of travel (ALPS).
IN1 = PWM, IN2 = 0: knob moves down. IN1 = IN2 = 1: brake.

MCU STM32C051F6P6

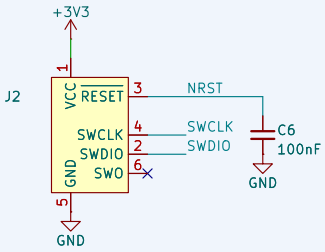
I2C pull-ups: on the host side (2.2k-4.7k to the bus logic level; SDA/SCL pins are 5 V tolerant, 3.3 V or 5 V bus OK)



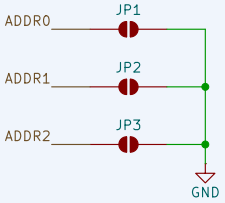
Status LED



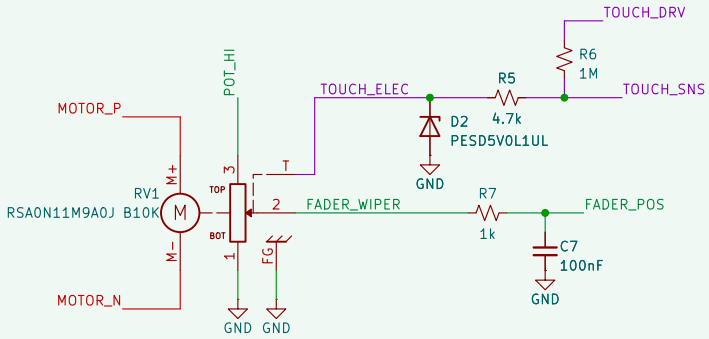
SWD (Tag-Connect TC2030-NL pads)



I2C address select: open = 1, bridged = 0



FADER: POSITION, TOUCH, MOTOR (ALPS RSA0N11M9A0J)



Touch: conductive knob -> lever -> T. Sense by RC charge time (PA1 drives via 1M, TIM2 capture on PA0) or by ADC charge sharing on PA0.

Position: track powered from POT_HI (PA2 high), wiper read on PA3 (ratiometric). End 1 (bottom of travel) = GND, so FADER_POS rises toward the top.

FG = fader frame lugs: soldered for retention, tied to GND.

Rev B: STM32C051F6P6 MCU, cost-reduced parts (Farnell LV), 4-layer board.
Fader: ALPS RSA0N11M9A0J (clone B0H45B29YK). 5 V + I2C on JST GH 4-pin.

Sheet: /
File: E0001_fader_driver.kicad_sch

Title: E0001 Motorized fader driver board

Size: A4	Date: 2026-10-07	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 1/1